

EN401 — Handout 6 Results & Analysis Report

Clean Decarbonization Transition & Seasonal Hydrological Dynamics

1. Solver Optimization Metrics

- **Status:** OPTIMAL | **Objective Value (NPV):** \$8,664.56 (-55.2% reduction vs. HO5)
- **Problem Scale:** 2,162 Rows, 1,440 Columns, 12,960 Non-Zeros | **Runtime:** < 0.12s

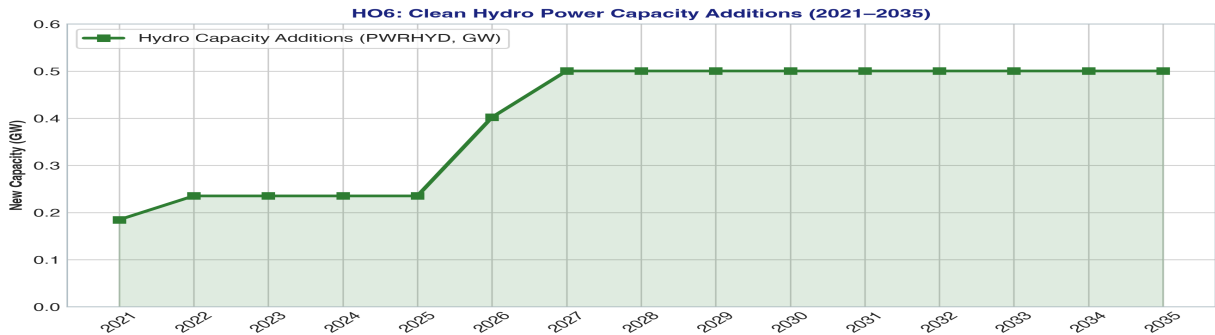


Figure: Hydro generation capacity additions reaching 4.85 GW by 2035.

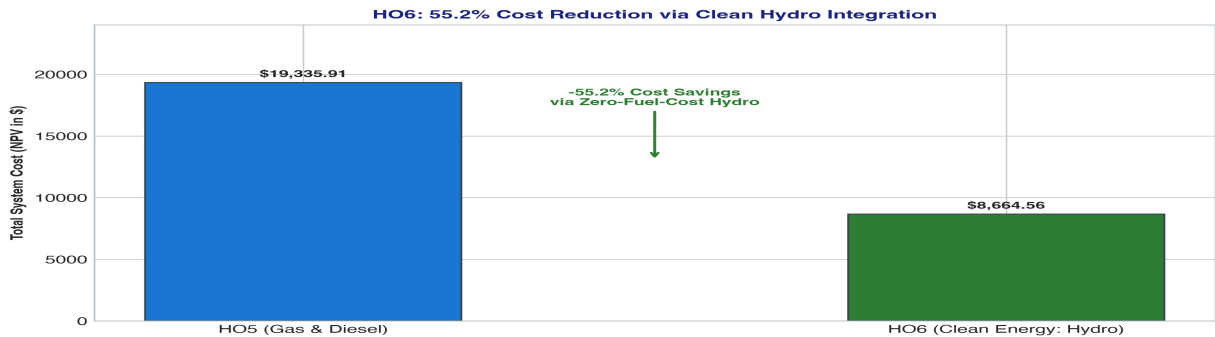


Figure: 55.2% additional cost savings achieved through hydro decarbonization.

2. Decarbonization Trajectory Data Table

Metric	2021	2023	2025	2027	2029	2031	2033	2035
Hydro Additions (GW)	0.185	0.235	0.235	0.500	0.500	0.500	0.500	0.500
Cumulative Hydro (GW)	0.185	0.655	1.126	2.126	3.127	4.128	4.628	4.850
Hydro Resource (PJ)	3.79	4.82	4.82	10.26	10.26	10.26	10.26	10.26
Gas Mining (PJ)	53.55	10.65	5.95	0.00	0.00	0.00	0.00	0.00

3. Seasonal Hydrology Insights

Hydro delivers 88% of generation. Gas plants operate exclusively during dry day hours (DD) to balance water deficits.